

Project Planning Proposal: Explainable CLV Ranking and Uplift-Aware Customer Targeting Across Retail Contexts

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Abstract

This proposal describes a ten-week Phase 2 Data Science project for ADY201m that extends our Phase 1 Customer Lifetime Value (CLV) framework from a single online-retail dataset to a multi-context study that connects *value* with *persuadability*. Phase 1 answered *who is valuable*; Phase 2 asks *who is worth targeting*. We re-estimate an explainable Hurdle–Semantic CLV model across UK online retail and US grocery contexts, then estimate *uplift* (incremental treatment effect) on a randomised supermarket SMS campaign, and finally combine the two into a *value-adjusted uplift* targeting rule. Following the mandated PBL + RBL framework, the plan covers a full pipeline (ingestion, cleaning, EDA, feature engineering, modelling, evaluation) in Python with supporting SQL analytics, more than five predictive models compared against published baselines, an interactive targeting dashboard, and a complete AI Audit Log demonstrating the *Human Delta*. A central design commitment is causal cleanliness: only the randomised X5 campaign is used for causal uplift, while coupon redemption is explicitly excluded as a treatment.

Keywords: Customer Lifetime Value, Uplift Modelling, RFM, Hurdle Model, Explainability, Conformal Prediction, Cross-Context Generalisation, Campaign Targeting

1 Project Name and Requirements

Project name: *Explainable CLV Ranking and Uplift-Aware Customer Targeting Across Retail Contexts – From “who is valuable” to “who is worth targeting”*.

ADY201m requirements addressed: a complete pipeline (Data → Cleaning → EDA → Features → Modelling → Evaluation); more than five predictive models compared against published baselines (eight CLV regressors plus six uplift learners); SQL analytics integrated with Python; an interactive targeting dashboard (Plotly Dash); a full AI Audit Log with Human Delta and at least three documented AI hallucinations or oversimplifications; a 10–12 page Final Report typeset in the Springer Nature single-column LaTeX template; and complete reproducibility (pinned environment and a single `reproduce_all` script).

2 Business Understanding and Analytic Approach

Marketing budgets are finite, yet value-based targeting alone can be wasteful. A high-CLV customer may purchase *regardless* of any campaign (a “sure buyer”), so retention dollars spent on them yield little incremental return. Conversely, a mid-value customer who only re-purchases when prompted is precisely the “persuadable” a campaign should reach. The business problem is therefore to identify customers who are *both valuable and incrementally responsive*, and to do so in a way that transfers across retail contexts and remains explainable to marketers.

The analytic approach is structured as four progressively prescriptive layers: *Descriptive* (RFM, cohort activity, cross-context dataset profiling); *Diagnostic* (SQL segmentation; statistical tests on CLV between segments; treatment/control balance audit); *Predictive* (CLV regression and uplift estimation); and *Prescriptive* (a dashboard ranking customers by value-adjusted uplift for activation).

The roles of CLV, Uplift, and Value-adjusted Uplift are complementary, not interchangeable: (i) **CLV** is the predictive value lens, a regression label equal to future revenue per customer; (ii) **Uplift** is the causal lens, the incremental effect of a randomised campaign; (iii) **Value-adjusted uplift** ($\hat{\tau}_i \cdot \hat{V}_i$) is the decision lens that fuses the two for budget-constrained activation. RQ1–RQ2 exercise CLV, RQ3 exercises uplift, and RQ4 exercises their fusion.

3 Research Questions (RBL)

- **RQ1.** How robust is the explainable Hurdle–Semantic CLV framework when re-estimated across UK online retail and US grocery contexts? *Method:* re-train the same pipeline on both datasets; compare Normalized Gini and Revenue Capture@10% with bootstrap confidence intervals.
- **RQ2.** Do semantic product profiles and behavioural sequence features add incremental value beyond RFM for CLV ranking, calibration, and interpretation? *Method:* feature-group ablation with paired bootstrap tests; SHAP attribution per Hurdle stage; conformalized quantile regression coverage.

- **RQ3.** On campaign data with treatment/control labels, does uplift-aware targeting outperform CLV-only, RFM-only, and response-model targeting? *Method:* S/T/X-Learners and class transformation vs. response and random baselines; Qini AUC, AUUC, uplift@K.
- **RQ4.** Can combining predicted value with uplift improve simulated campaign efficiency, and what transferable design principles follow for budget-constrained markets? *Method:* value-adjusted uplift policy; profit / incremental-revenue simulation on the randomised test set with bootstrap CIs.

4 Data Collection, Understanding, and Preparation

Three datasets with deliberately separated roles. (1) *Online Retail II* (UK e-commerce, Phase 1 baseline, read-only reuse): ~ 0.8 M cleaned transactions, 5,038 observation customers. (2) *Dunnhumby Complete Journey* (US grocery): 2,500 households, ~ 2.6 M transaction lines over 102 weeks; external CLV validation. (3) *X5 RetailHero* (randomised SMS campaign): 200,039 labelled clients with `treatment_flg/target` and 45.7M pre-communication purchase lines; the sole source of causal uplift.

Causal-cleanliness commitment. Dunnhumby coupon redemption is a post-campaign outcome, not a clean treatment assignment; using it for causal uplift would invite selection bias. We therefore restrict uplift to the randomised X5 design (audited treatment ratio 0.500, raw uplift +3.3 percentage points with a positive 95% confidence interval) and use Dunnhumby only for CLV validation.

Preparation pipeline. A shared normaliser maps each dataset to a common transaction schema (customer, basket, date, product, quantity, revenue); observation and prediction windows are split without leakage; the CLV label is future-window revenue $Y_i = \sum_t \text{Revenue}_{i,t}$; and X5 features are aggregated from pre-communication purchases only.

5 Proposed Methodology and Tools

The methodology has four modules. **Module A (Hurdle CLV)** models value as $\widehat{CLV}_i = p_i \cdot m_i$ with incidence $p_i = P(Y_i > 0 \mid X_i)$ and magnitude $m_i = E[Y_i \mid Y_i > 0, X_i]$ using gradient boosting and a lognormal correction. **Module B (Semantic profiling)** builds PPMI product co-occurrence embeddings via truncated SVD and revenue-weighted customer taste vectors. **Module C (Uplift)** estimates $\tau_i = E[Y_i(1) - Y_i(0) \mid X_i]$ with S/T/X-Learners and class transformation. **Module D (Value-adjusted uplift)** scores customers by $\hat{\tau}_i \cdot \hat{V}_i$, where $\hat{V}_i$ is a historical-spend value *proxy*.

Tools. `pandas`, `numpy`, `scipy`, `scikit-learn`, `LightGBM`, `XGBoost`, `SHAP`, `scikit-uplift`, `matplotlib`; SQL via DuckDB; dashboard via Plotly Dash. Headline metrics: Normalized Gini and Revenue Capture@10% (CLV); Qini AUC and uplift@K (uplift); profit@K (policy). Uncertainty via the percentile bootstrap and conformalized quantile regression.

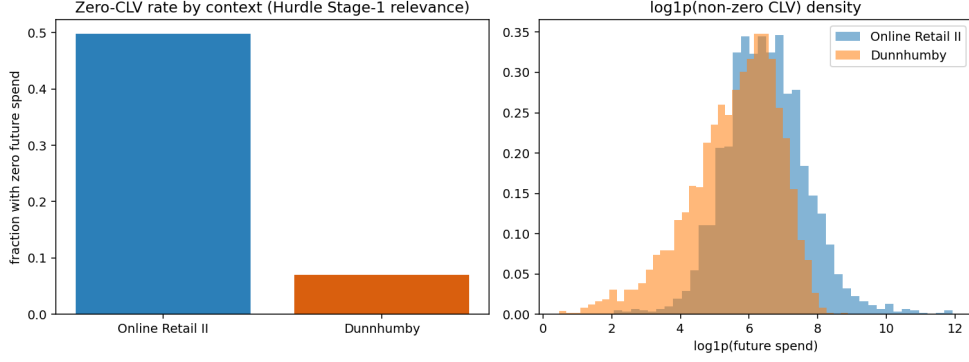


Fig. 1 Illustrative cross-context structure: e-commerce is zero-inflated and heavy-tailed, grocery is near fully active and light-tailed – motivating the RQ1 robustness test.

6 Data Analysis with SQL

Seven DuckDB queries support the four RQs across the datasets:

- `q1_rfm_per_customer` — R/F/M aggregation per customer/household (RQ1/RQ2);
- `q2_rfm_segments` — quintile RFM scores and named segments (RQ1);
- `q3_cohort_activity` — first-active-period $\times$ activity matrix (RQ1);
- `q4_zero_rate_by_context` — future-spend zero-rate and skew per context (RQ1);
- `q5_treatment_balance` — X5 treatment/control counts and target rates (RQ3);
- `q6_top_value_customers` — top-decile value customers per context (RQ4);
- `q7_uplift_decile_overlap` — overlap of top-response vs top-uplift deciles (RQ4).

Deliverables: one self-contained SQL notebook, seven CSV outputs, and a paragraph linking each query to an RQ.

7 Data Analysis with Python

Eleven reproducible Python modules (p2.01–p2.11): dataset audits; a shared feature pipeline (RFM, behavioural, sequence, and semantic-graph features); cross-context comparison; CLV benchmark and ablation; SHAP and CQR; X5 uplift feature engineering; uplift models; the value-adjusted policy comparison; bootstrap robustness; and an automatic LaTeX-table generator that keeps the report numbers synchronised with the result files. Required libraries are listed in a pinned `requirements.txt`.

8 Data Visualisation

Core figures: cross-context dataset structure (zero-rate and tail, Fig. 1); revenue-capture and decile-calibration curves per context; stage-wise SHAP summaries; conformal interval widths; X5 Qini curves; and targeting profit/incremental-revenue by budget. Each figure is generated by a script and saved under `figures/` for direct inclusion in the report.

9 Predictive Modelling ($\geq$ Five Models)

CLV regressors (eight). Mean and Monetary baselines, an RFM-score baseline, Ridge, Random Forest, XGBoost, LightGBM, and the two-stage *Hurdle* model (proposed). Metrics: Normalized Gini, Revenue Capture@10/20%, Precision@10%, Spearman, MAE/RMSE, decile calibration. **Uplift learners (six).** Random and Response baselines, S-Learner, T-Learner, class transformation, and X-Learner. Metrics: Qini AUC, AUUC, uplift@10/20/30%. Significance is assessed by paired bootstrap differences (e.g. *Hurdle* vs. baseline; best uplift learner vs. response model). This comfortably exceeds the five-model requirement and compares against published baselines (BG/NBD-style monetary models for CLV; response and random models for uplift).

10 Data Analysis with a Tool (Dashboard)

An interactive Plotly Dash application with four pages: (1) cross-context CLV overview; (2) per-customer CLV ranking with conformal intervals; (3) uplift Qini explorer; and (4) a value-adjusted targeting page that ranks the top- K persuadable high-value customers under a budget slider, with the simulated profit shown live. The dashboard consumes the CSV outputs of the Python pipeline.

11 Ten-Week Timeline and Deliverables

- **W1** Dataset audits + feasibility gating (treatment balance, zero-rate, skew).
- **W2** Shared feature engineering (RFM, behavioural, sequence, semantic).
- **W3** CLV window design (week-level for grocery).
- **W4** Cross-context CLV benchmark + ablation.
- **W5** Explainability (SHAP per stage) + uncertainty (CQR).
- **W6** X5 uplift feature engineering + baselines.
- **W7** Advanced uplift learners (X-Learner, class transformation).
- **W8** Value-adjusted uplift policy + profit simulation.
- **W9** Robustness (bootstrap CIs) + transferable insights.
- **W10** Report writing + reproducibility package.

Deliverables: cleaned data and shared feature tables; result CSVs and figures; the AI Audit Log; a 10–12 page Springer report; a reproducibility bundle (`requirements.txt`, `reproduce_all.sh`); and the dashboard.

12 Risks, Mitigations, and AI Reflection

Risk 1 – causal misuse. Treating coupon redemption as a campaign cause would bias uplift. *Mitigation:* restrict causal estimation to the randomised X5 design. **Risk 2 – data acquisition.** The X5 purchase history is large (~ 670 MB) and the link unreliable. *Mitigation:* a resumable, integrity-checked downloader; manual fallback. **Risk 3 – overclaiming.** Profit differences between policies may be within noise. *Mitigation:* bootstrap confidence intervals and explicit reporting of non-significant

claims. **Risk 4 – leakage / isolation.** Phase 2 must not corrupt Phase 1 results. *Mitigation:* a separate project tree with a write-guard that refuses paths outside it. **AI reflection (Human Delta).** AI assistance is used for ideation and drafting, but every quantitative claim is recomputed from data, every cited reference is verified, and at least three AI oversimplifications (e.g. “response model is a strong targeting baseline”; “coupon redemption is a valid treatment”; “semantic features must improve ranking”) are documented and corrected in the AI Audit Log with the reasoning behind each human decision.